

SmoIVLM

Redefining Small and Efficient Multimodal Models

Hugging Face & Stanford University

Compact Vision-Language Models · Sub-1GB Inference · On-Device AI

The Edge Deployment Problem

State-of-the-art VLMs require massive compute and memory, making mobile and edge deployment impractical.

Idefics-80B

80B params

~32GB+

300× larger

Molmo-A1B-7B

7B params

27.7GB

35× larger

InternVL2-2B

2B params

~10GB

10× larger

Goal: Achieve competitive performance with <1GB GPU RAM — enabling true on-device inference.

SmolVLM Architecture Pipeline

1

Image Splitting

Input image is divided into sub-images (tiles) for efficient processing

2

SigLIP Vision Encoder

Encodes each tile with a lightweight SigLIP-B/16 (93M params)

3

Pixel Shuffle

Space-to-depth transform reduces tokens $16\times$ at $r=4$

4

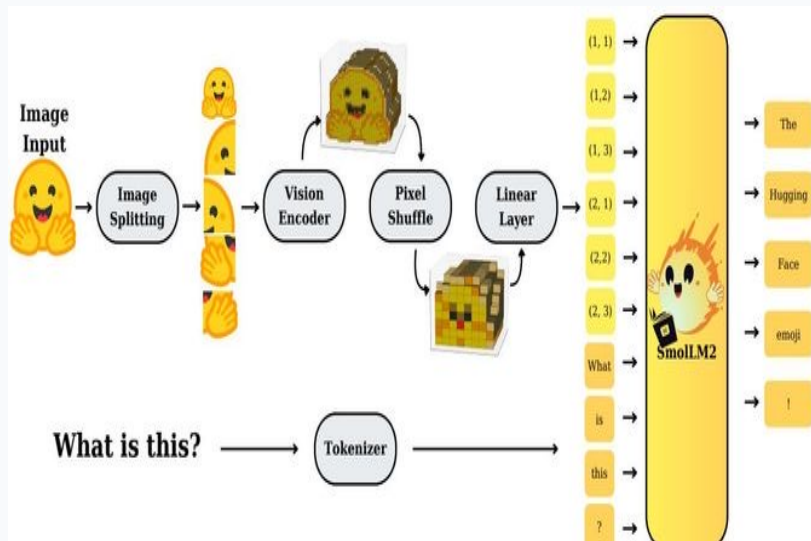
Linear Projection

Maps visual features into SmolLM2's embedding space

5

SmolLM2 Backbone

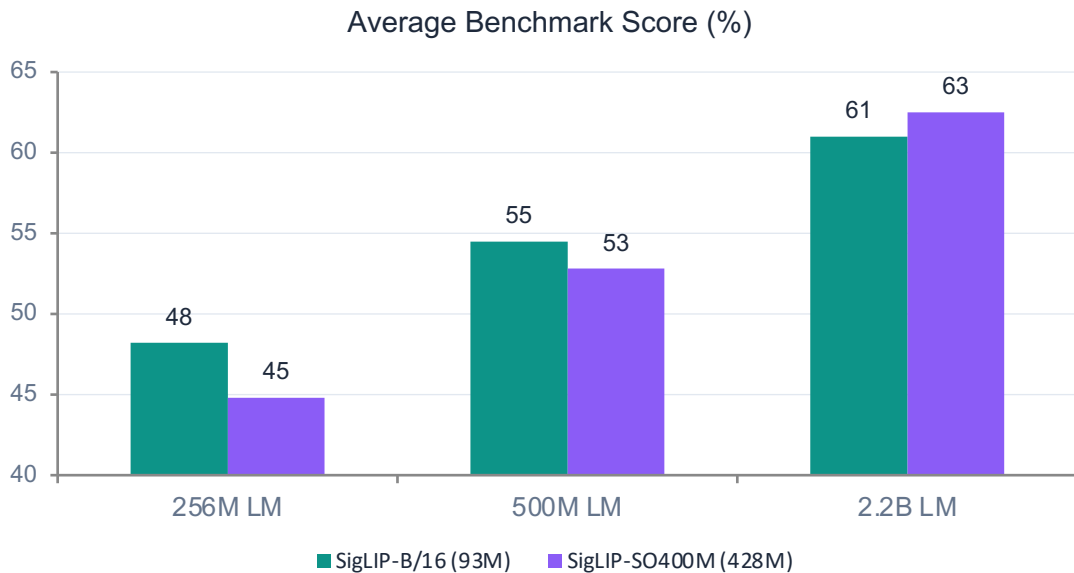
Causal language model (135M / 360M / 1.7B params)



Video: Frames are sampled and processed identically to images through the same pipeline.

Finding 1 — Balanced Encoder-LM Compute Allocation

Smaller vision encoders outperform larger ones for compact VLMs



Key Insight

For LM < 1B:

93M SigLIP-B/16 pairs best

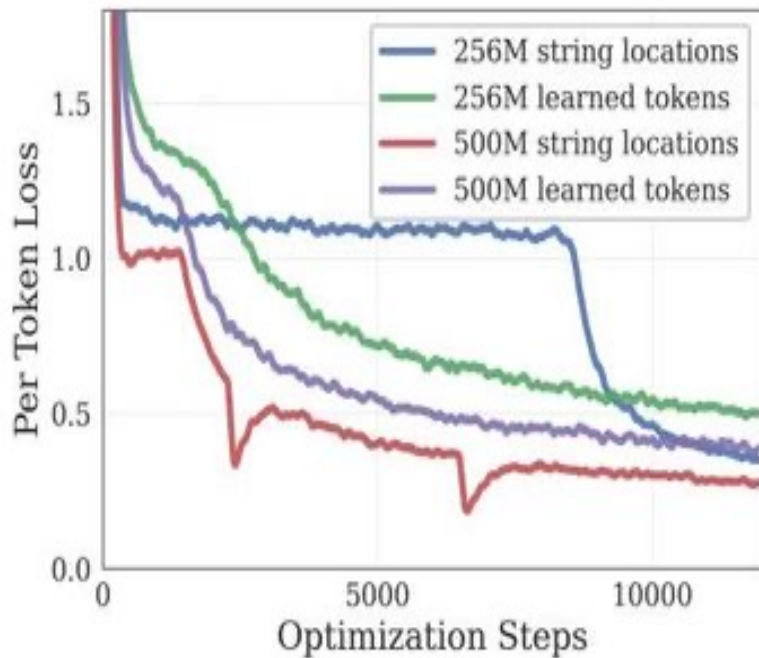
Only at 1.7B+ LM scale:

Does SO400M justify its 10% parameter overhead

Recommendation:

Balance encoder vs. LM size — don't over-invest in vision encoder for small VLMs

Finding 3 — Pixel Shuffle: 16× Token Reduction



How Pixel Shuffle Works

Space-to-depth transform rearranges spatial features into channel dimension

Factor r groups pixels into blocks, stacking channels

Token count reduces by r^2 (e.g., $r=4 \rightarrow 16\times$ fewer tokens)

Token Reduction by Compression Factor

$r = 2$

4× token reduction

Better for larger VLMs

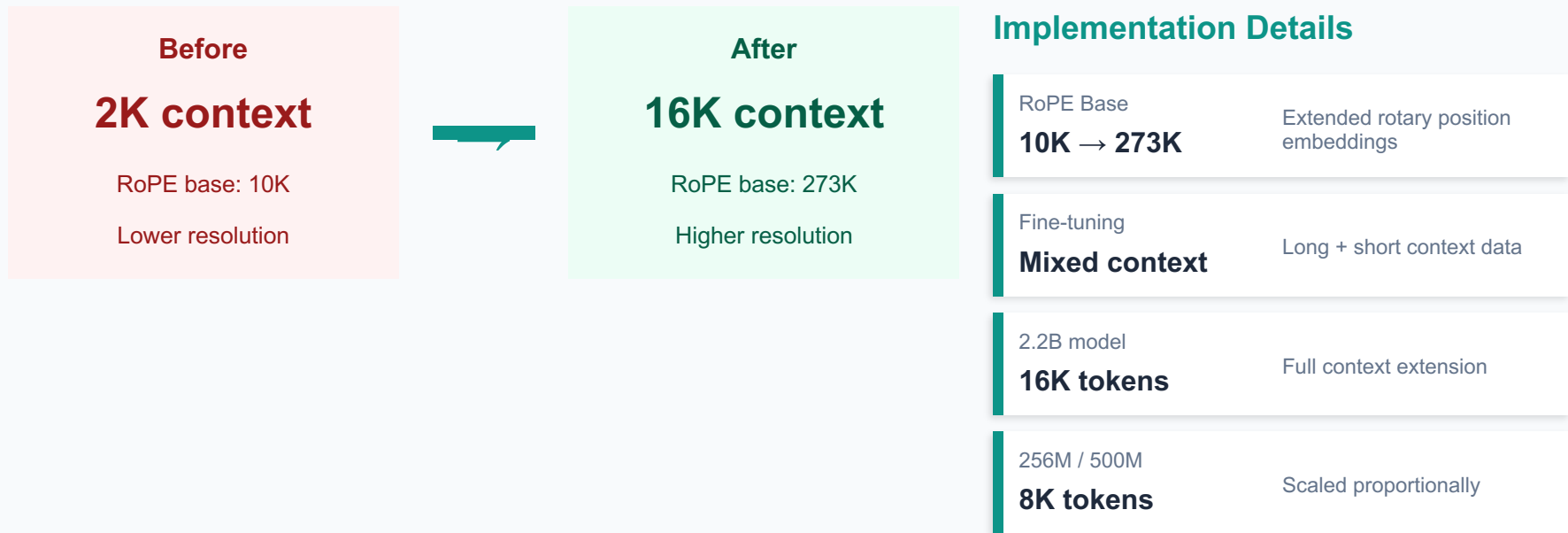
$r = 4$

16× token reduction

Optimal for small VLMs

Aggressive compression eases attention overhead and enables longer-context modeling for small VLMs.

Finding 2 — Extended Context: 2K → 16K Tokens



Impact: Significant benchmark gains — higher resolution images unlock more visual reasoning capacity.

Benchmark Performance: Competitive at Sub-1GB RAM

SmolVLM 256M

44.0%

0.8 GB RAM

SmolVLM 500M

51.0%

1.2 GB RAM

SmolVLM 2.2B

59.8%

4.9 GB RAM

Capability	Benchmark	256M	500M	2.2B	Efficient OS
Single-Image	OCRBench	52.6%	61.0%	72.9%	54.7% (Molmo)
Single-Image	AI2D	46.4%	59.2%	70.0%	71.0% (Molmo)
Single-Image	ChartQA	55.6%	62.8%	68.7%	48.0% (Molmo)
Single-Image	TextVQA	50.2%	60.2%	73.0%	61.5% (Molmo)
Single-Image	DocVQA	58.3%	70.5%	80.0%	77.7% (Molmo)
Single-Image	ScienceQA	73.8%	80.0%	89.6%	87.5% (Molmo)
Multi-task	MMMU	29.0%	33.7%	42.0%	33.9% (Molmo)
Multi-task	MathVista	35.9%	40.1%	51.5%	37.6% (Molmo)
Multi-task	MMStar	34.6%	38.3%	46.0%	43.1% (Molmo)
Video	Video-MME	33.7%	42.2%	52.1%	45.0% (InternVL2)
Video	MLVU	40.6%	47.3%	55.2%	48.2% (InternVL2)
Average	All 11 Benchmarks	44.0%	51.0%	59.8%	—

RAM: SmolVLM-256M: 0.8GB | 500M: 1.2GB | 2.2B: 4.9GB vs. Molmo-A1B-7B: 27.7GB (35× more)

Real-Time Inference on Apple Silicon



HuggingSnap iOS App — Local Inference

Photos and videos processed entirely on-device — no cloud dependency.

Pro M1 Max 64GB

~80 tok/s

batch 1

Pro M1 Max 64GB (batch 64)

~1000 tok/s

batch 64

MacBook Air M3 16GB

~55 tok/s

batch 1

Throughput scales linearly with batch size for SmolVLM-256M

80tok/s

on M1 Max Pro

1000tok/s

batch 64 on M1 Max

0.8GB

GPU RAM at batch 1

Design Principles for Efficient Multimodal Models

01 Balance Encoder-LM Compute

Don't overspend on vision encoder for small VLMs. 93M SigLIP-B/16 outperforms 428M SO400M for LM < 1B.

02 Extend Context Length

RoPE base 10K → 273K enables 16K context. Higher resolution unlocks more visual reasoning capacity.

03 Aggressive Token Compression

$r=4$ (16× reduction) is optimal for small VLMs — eases attention overhead and improves long-context modeling.

04 Sub-1GB Inference Achievable

SmolVLM-256M at 0.8GB surpasses the 300× larger Llafix-80B. Architectural efficiency > raw scale.

05 Fully Open-Source

All weights, datasets, code, and HuggingSnap mobile app are released to promote reproducibility.